

LIBRARY MANAGEMENT SYSTEM

PROJECT REPORT

DEVELOPED USING: Python and Mysql

Submitted By:

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INTRODUCTION

THE **LIBRARY MANAGEMENT SYSTEM** IS A CONSOLE-BASED APPLICATION DEVELOPED USING **PYTHON AND MYSQL**. THE MAIN PURPOSE OF THIS PROJECT IS TO MANAGE COMMON LIBRARY ACTIVITIES SUCH AS MAINTAINING BOOK RECORDS, REGISTERING MEMBERS, ISSUING BOOKS, AND PROCESSING BOOK RETURNS.

THE SYSTEM STORES LIBRARY INFORMATION IN A MYSQL DATABASE AND PROVIDES A SIMPLE MENU-DRIVEN INTERFACE FOR USERS. IT REDUCES MANUAL RECORD KEEPING AND MAKES LIBRARY OPERATIONS EASIER TO MANAGE.

THE PROJECT DEMONSTRATES THE PRACTICAL USE OF PYTHON PROGRAMMING, DATABASE CONNECTIVITY, SQL QUERIES, RELATIONAL DATABASE CONCEPTS, EXCEPTION HANDLING, AND TRANSACTION MANAGEMENT.

PROBLEM STATEMENT

MANAGING LIBRARY RECORDS MANUALLY CAN BE TIME-CONSUMING AND MAY LEAD TO ERRORS IN MAINTAINING BOOK AND MEMBER INFORMATION.

THE MAIN PROBLEMS INCLUDE:

- DIFFICULTY IN MAINTAINING BOOK RECORDS.
- DIFFICULTY IN TRACKING REGISTERED MEMBERS.
- MANUAL TRACKING OF ISSUED AND RETURNED BOOKS.
- DIFFICULTY IN MAINTAINING AVAILABLE BOOK QUANTITIES.
- POSSIBILITY OF DUPLICATE OR INCORRECT RECORDS.
- TIME-CONSUMING SEARCHING OF BOOKS AND MEMBERS.

THE PROPOSED LIBRARY MANAGEMENT SYSTEM PROVIDES A COMPUTERIZED SOLUTION FOR MANAGING THESE ACTIVITIES USING PYTHON AND MYSQL.

FUNCTIONAL REQUIREMENTS

THE SYSTEM PROVIDES THE FOLLOWING FUNCTIONAL REQUIREMENTS:

4.1 BOOK MANAGEMENT

- ADD NEW BOOKS.

- VIEW ALL BOOKS.
- SEARCH BOOKS BY TITLE OR AUTHOR.
- DELETE BOOKS.
- MAINTAIN BOOK QUANTITIES.

4.2 MEMBER MANAGEMENT

- REGISTER NEW MEMBERS.
- STORE MEMBER NAMES AND PHONE NUMBERS.
- VIEW REGISTERED MEMBERS.

4.3 BOOK ISSUE MANAGEMENT

- ISSUE BOOKS TO REGISTERED MEMBERS.
- CHECK WHETHER A BOOK IS AVAILABLE.
- CHECK WHETHER THE MEMBER EXISTS.
- PREVENT THE SAME MEMBER FROM RECEIVING THE SAME BOOK TWICE.
- AUTOMATICALLY DECREASE BOOK QUANTITY AFTER ISSUING.

4.4 BOOK RETURN MANAGEMENT

- RETURN AN ISSUED BOOK USING ITS ISSUE ID.
- RECORD THE RETURN DATE.
- AUTOMATICALLY INCREASE BOOK QUANTITY AFTER RETURNING.

4.5 ISSUE RECORDS

- VIEW ISSUED BOOK RECORDS.
 - DISPLAY ISSUE DATES.
 - DISPLAY RETURN DATES.
 - IDENTIFY BOOKS THAT HAVE NOT YET BEEN RETURNED.
-

5. NON-FUNCTIONAL REQUIREMENTS

5.1 USABILITY

THE APPLICATION PROVIDES A SIMPLE MENU-DRIVEN INTERFACE THAT IS EASY TO UNDERSTAND AND OPERATE.

5.2 RELIABILITY

THE SYSTEM USES DATABASE TRANSACTIONS AND ERROR HANDLING TO MAINTAIN RELIABLE RECORDS.

5.3 PERFORMANCE

THE APPLICATION PERFORMS BASIC DATABASE OPERATIONS QUICKLY FOR A SMALL OR MEDIUM-SIZED LIBRARY.

5.4 SECURITY

PARAMETERIZED SQL QUERIES ARE USED WHEN INSERTING AND SEARCHING DATA, REDUCING THE RISK OF SQL INJECTION THROUGH USER INPUT.

5.5 MAINTAINABILITY

THE PROGRAM IS DIVIDED INTO SEPARATE FUNCTIONS FOR DIFFERENT OPERATIONS, MAKING IT EASIER TO UNDERSTAND AND MODIFY.

5.6 DATA INTEGRITY

PRIMARY KEYS, FOREIGN KEYS, AND DATABASE CONSTRAINTS ARE USED TO MAINTAIN RELATIONSHIPS AND CONSISTENCY BETWEEN RECORDS.

6. SYSTEM ARCHITECTURE

THE SYSTEM FOLLOWS A SIMPLE THREE-LAYER STRUCTURE:

1. USER INTERFACE LAYER

THE USER INTERACTS WITH THE APPLICATION THROUGH THE COMMAND-LINE MENU.

2. APPLICATION LAYER

PYTHON HANDLES:

- USER INPUT
- VALIDATION
- BUSINESS LOGIC
- BOOK ISSUE/RETURN OPERATIONS
- ERROR HANDLING

3. DATABASE LAYER

MYSQL STORES:

- BOOK RECORDS
- MEMBER RECORDS
- ISSUE AND RETURN RECORDS

ARCHITECTURE FLOW

USER → PYTHON APPLICATION → MYSQL DATABASE

THE USER SELECTS AN OPERATION FROM THE MENU. PYTHON PROCESSES THE REQUEST AND EXECUTES THE REQUIRED SQL QUERY. MYSQL STORES OR RETRIEVES THE REQUESTED INFORMATION AND SENDS THE RESULT BACK TO THE PYTHON APPLICATION.

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DESIGN DIAGRAMS

7.1 USE CASE DIAGRAM

ACTORS

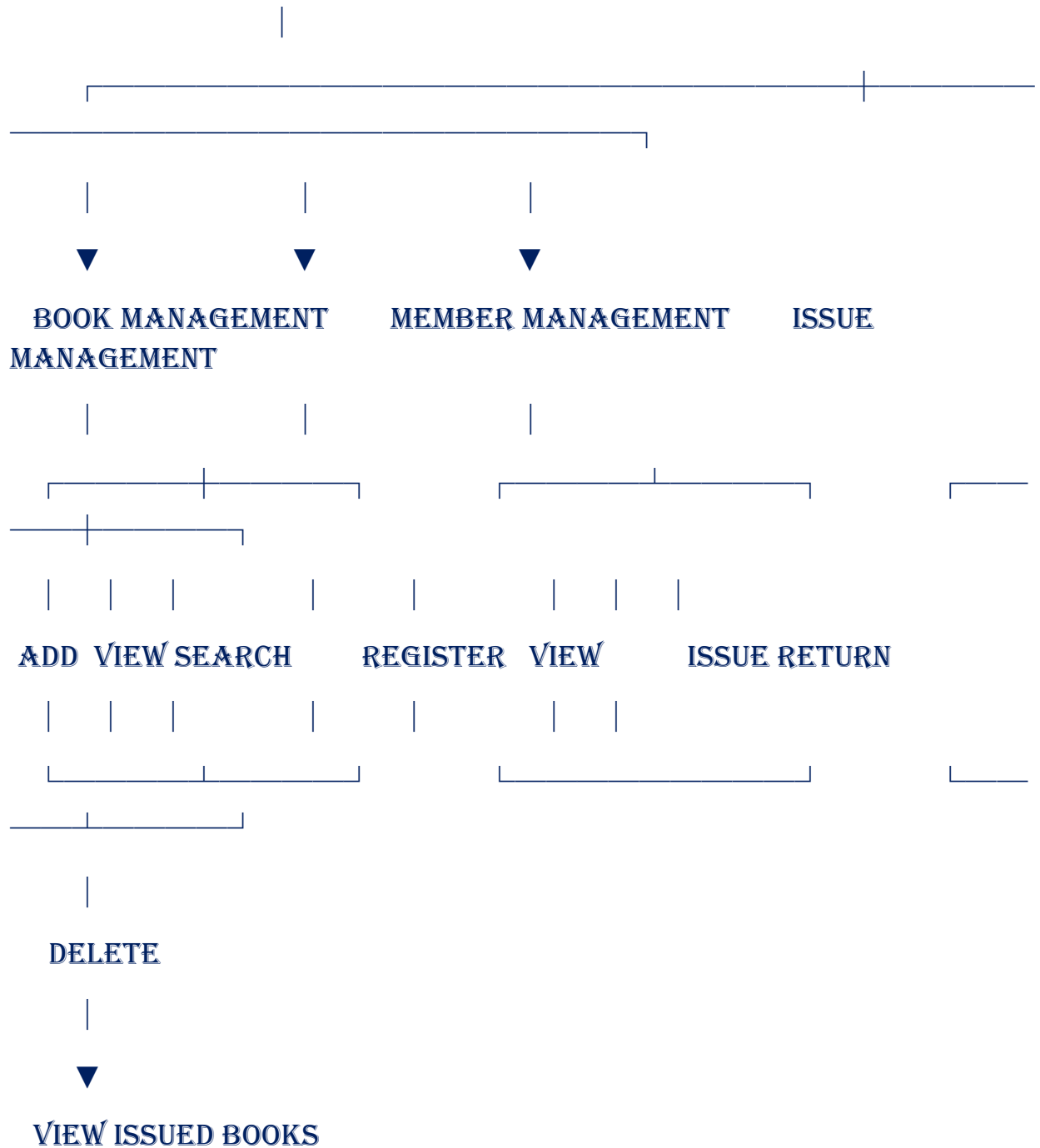
LIBRARY USER / ADMIN

MAIN USE CASES

- ADD BOOK
- VIEW BOOKS
- SEARCH BOOK
- DELETE BOOK
- REGISTER MEMBER
- VIEW MEMBERS
- ISSUE BOOK
- RETURN BOOK
- VIEW ISSUED BOOKS

DIAGRAM REPRESENTATION

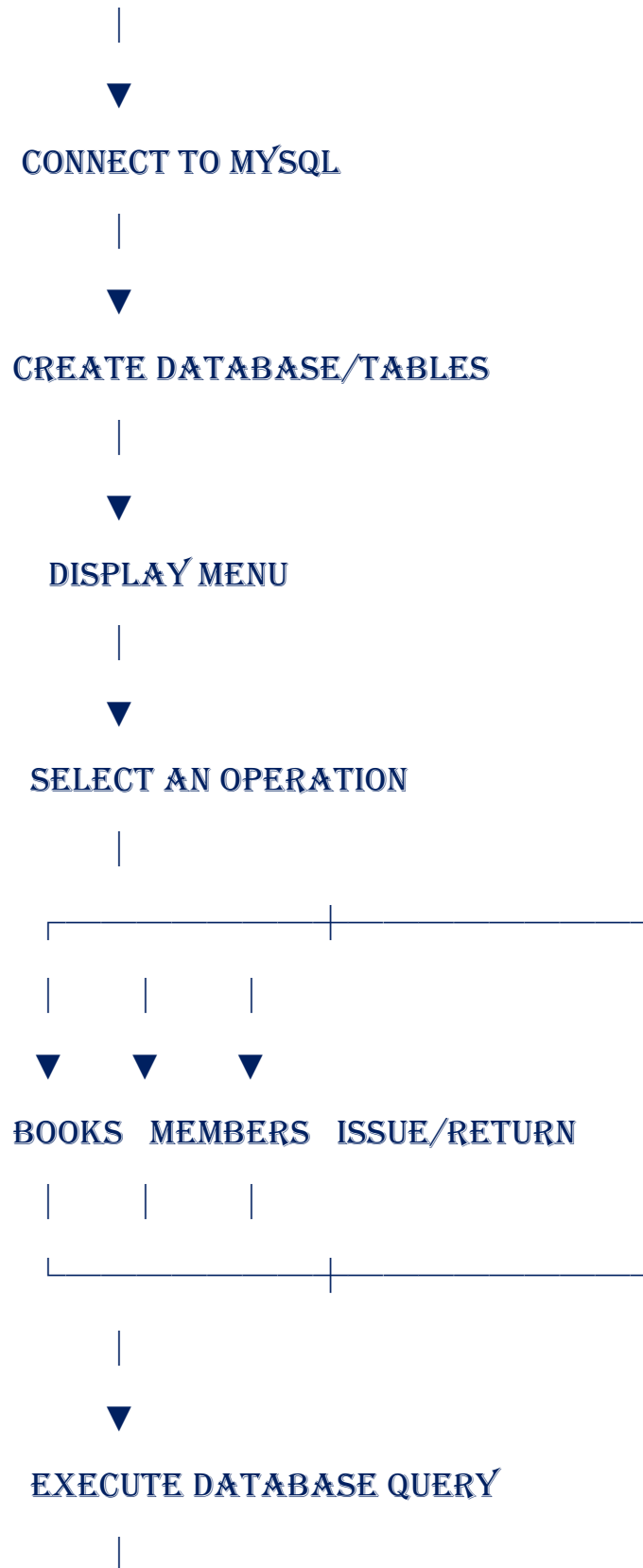


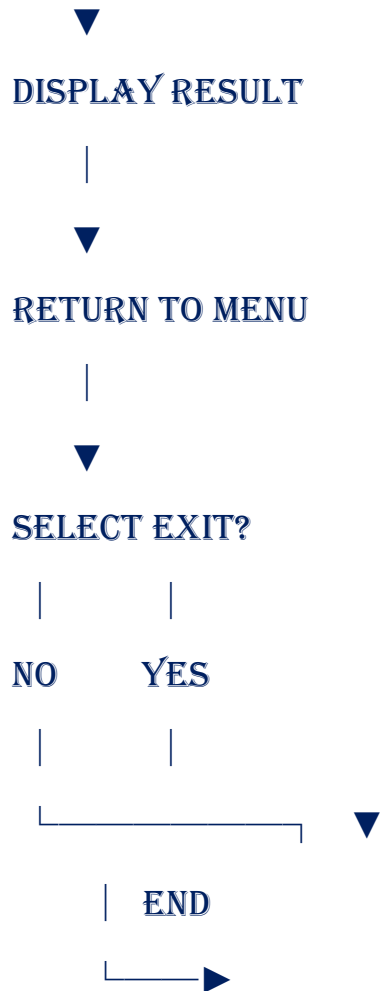


7.2 WORKFLOW DIAGRAM

THE GENERAL WORKFLOW OF THE APPLICATION IS:

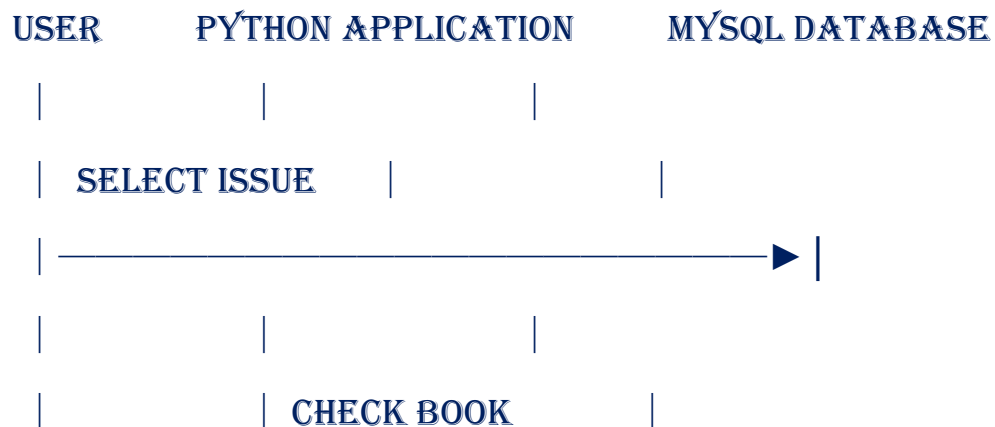
START

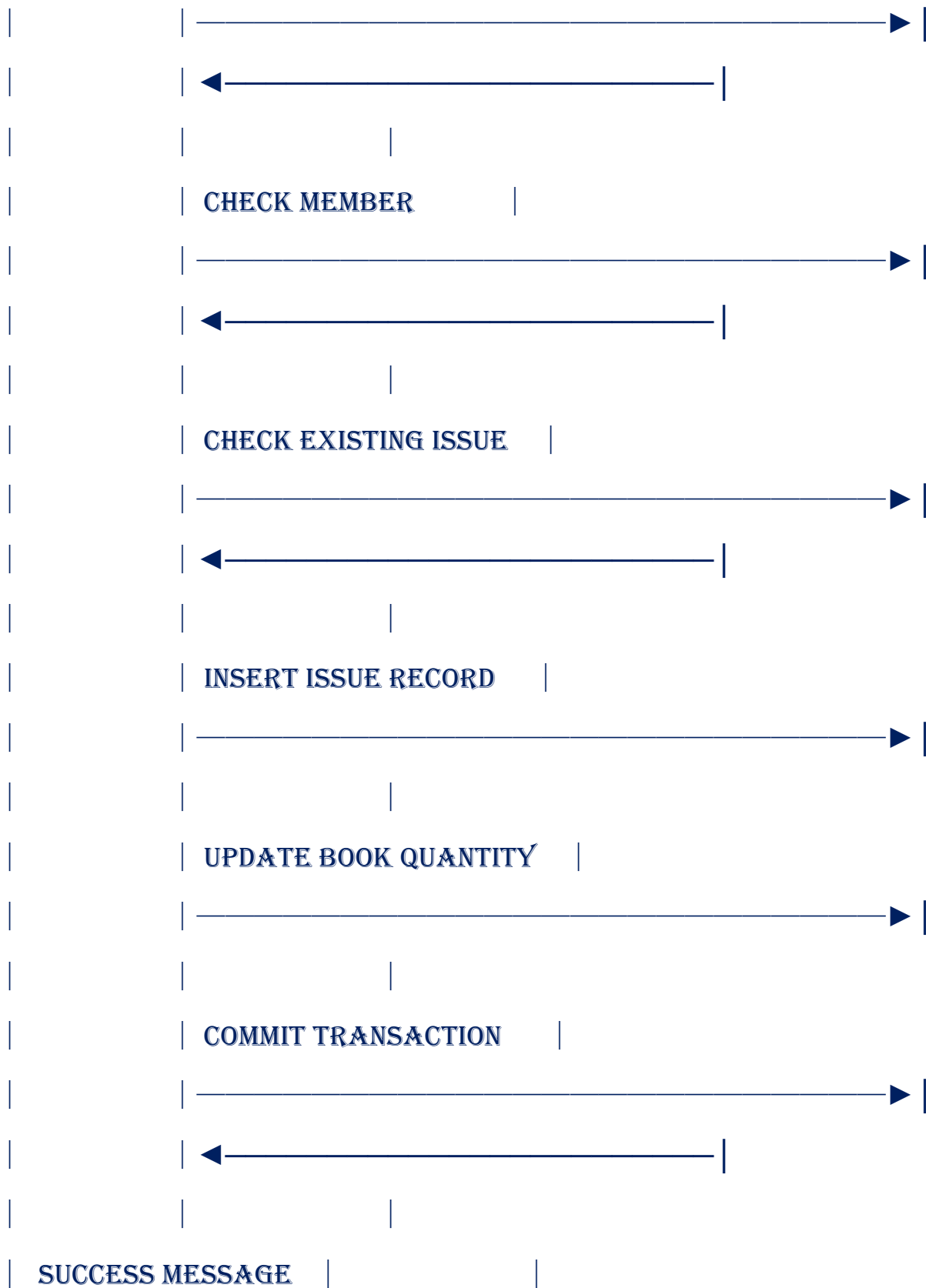




7.3 SEQUENCE DIAGRAM

EXAMPLE: ISSUE BOOK



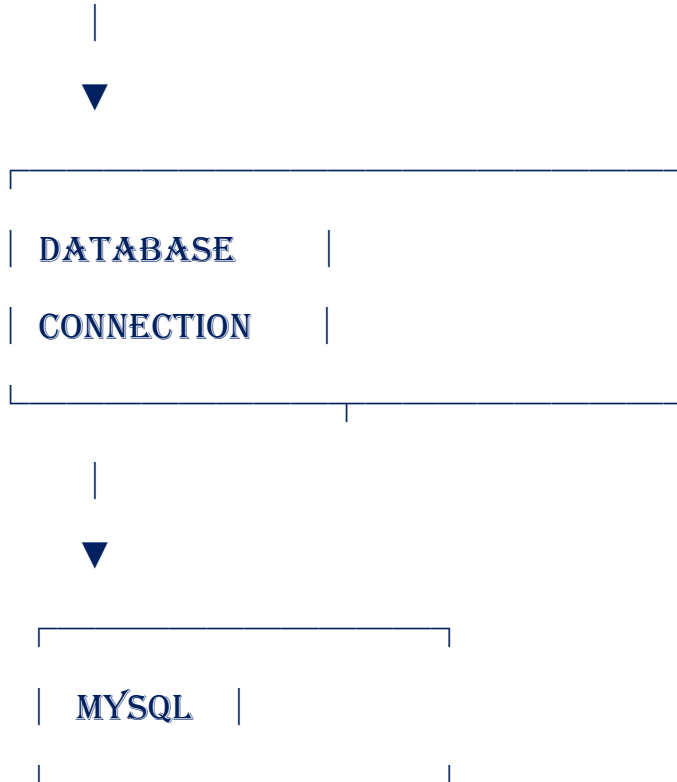




7.4 CLASS / COMPONENT DIAGRAM

THE APPLICATION IS ORGANIZED INTO FUNCTIONAL COMPONENTS.





MAJOR PYTHON COMPONENTS

- `CONNECT_DATABASE()` – CONNECTS THE APPLICATION TO MYSQL.
- `SETUP_DATABASE()` – CREATES THE DATABASE AND TABLES.
- `ADD_BOOK()` – ADDS BOOKS.
- `VIEW_BOOKS()` – DISPLAYS BOOKS.
- `SEARCH_BOOK()` – SEARCHES BOOKS.
- `DELETE_BOOK()` – DELETES BOOKS.
- `ADD_MEMBER()` – REGISTERS MEMBERS.
- `VIEW_MEMBERS()` – DISPLAYS MEMBERS.
- `ISSUE_BOOK()` – ISSUES BOOKS.
- `RETURN_BOOK()` – PROCESSES RETURNS.
- `VIEW_ISSUED_BOOKS()` – DISPLAYS ISSUE RECORDS.

- MAIN() – CONTROLS THE MAIN MENU AND APPLICATION FLOW.

7.5 ER DIAGRAM

THE DATABASE CONSISTS OF THREE MAIN ENTITIES:

BOOKS

- BOOK ID – PRIMARY KEY
- TITLE
- AUTHOR
- QUANTITY

MEMBERS

- MEMBER ID – PRIMARY KEY
- NAME
- PHONE

ISSUED BOOKS

- ISSUE ID – PRIMARY KEY
- BOOK ID – FOREIGN KEY
- MEMBER ID – FOREIGN KEY
- ISSUE DATE
- RETURN DATE

ER DIAGRAM REPRESENTATION



TITLE	
AUTHOR	
QUANTITY	

1	
N	

ISSUED_BOOKS	

PK ISSUE_ID	
FK BOOK_ID	
FK MEMBER_ID	
ISSUE_DATE	
RETURN_DATE	

N	
1	

		▼		
MEMBERS				
PK MEMBER_ID				
NAME				
PHONE				

A BOOK CAN HAVE MULTIPLE ISSUE RECORDS OVER TIME, AND A MEMBER CAN HAVE MULTIPLE ISSUE RECORDS.

DESIGN DECISION AND RATIONALE

PYTHON

PYTHON WAS SELECTED BECAUSE IT IS SIMPLE, READABLE, AND PROVIDES STRONG SUPPORT FOR DATABASE CONNECTIVITY.

MYSQL

MYSQL WAS SELECTED BECAUSE IT IS A RELIABLE RELATIONAL DATABASE SYSTEM AND IS SUITABLE FOR STORING STRUCTURED LIBRARY INFORMATION.

MENU-DRIVEN INTERFACE

A CONSOLE-BASED MENU WAS CHOSEN BECAUSE IT IS SIMPLE TO IMPLEMENT AND ALLOWS USERS TO EASILY SELECT OPERATIONS.

SEPARATE FUNCTIONS

EACH MAJOR OPERATION IS IMPLEMENTED AS A SEPARATE FUNCTION. THIS IMPROVES CODE READABILITY, ORGANIZATION, AND MAINTAINABILITY.

PRIMARY AND FOREIGN KEYS

PRIMARY KEYS UNIQUELY IDENTIFY RECORDS, WHILE FOREIGN KEYS MAINTAIN RELATIONSHIPS BETWEEN BOOKS, MEMBERS, AND ISSUE RECORDS.

TRANSACTIONS

DATABASE CHANGES DURING ISSUING AND RETURNING ARE COMMITTED TOGETHER. IF AN ERROR OCCURS, THE TRANSACTION CAN BE ROLLED BACK.

PARAMETERIZED QUERIES: SQL QUERIES ARE USED TO SAFELY HANDLE USER-PROVIDED VALUES.

IMPLEMENTATION DETAILS

THE APPLICATION IS IMPLEMENTED USING PYTHON AND MYSQL CONNECTOR.

DATABASE CONNECTION

THE PROGRAM ESTABLISHES A CONNECTION WITH THE LOCAL MYSQL SERVER.

DATABASE CREATION

THE APPLICATION CREATES THE LIBRARY_DB DATABASE IF IT DOES NOT ALREADY EXIST.

TABLE CREATION

THREE TABLES ARE CREATED:

- BOOKS**
- MEMBERS**
- ISSUED_BOOKS**

BOOK MANAGEMENT

THE SYSTEM ALLOWS USERS TO ADD, DISPLAY, SEARCH, AND DELETE BOOKS.

MEMBER MANAGEMENT

THE SYSTEM STORES AND DISPLAYS MEMBER INFORMATION.

ISSUE PROCESS

BEFORE ISSUING A BOOK, THE APPLICATION CHECKS:

- 1. WHETHER THE BOOK EXISTS.**
- 2. WHETHER THE BOOK IS AVAILABLE.**
- 3. WHETHER THE MEMBER EXISTS.**
- 4. WHETHER THE MEMBER ALREADY HAS THE SAME BOOK.**

AFTER SUCCESSFUL ISSUE, THE BOOK QUANTITY IS DECREASED.

RETURN PROCESS

WHEN A BOOK IS RETURNED:

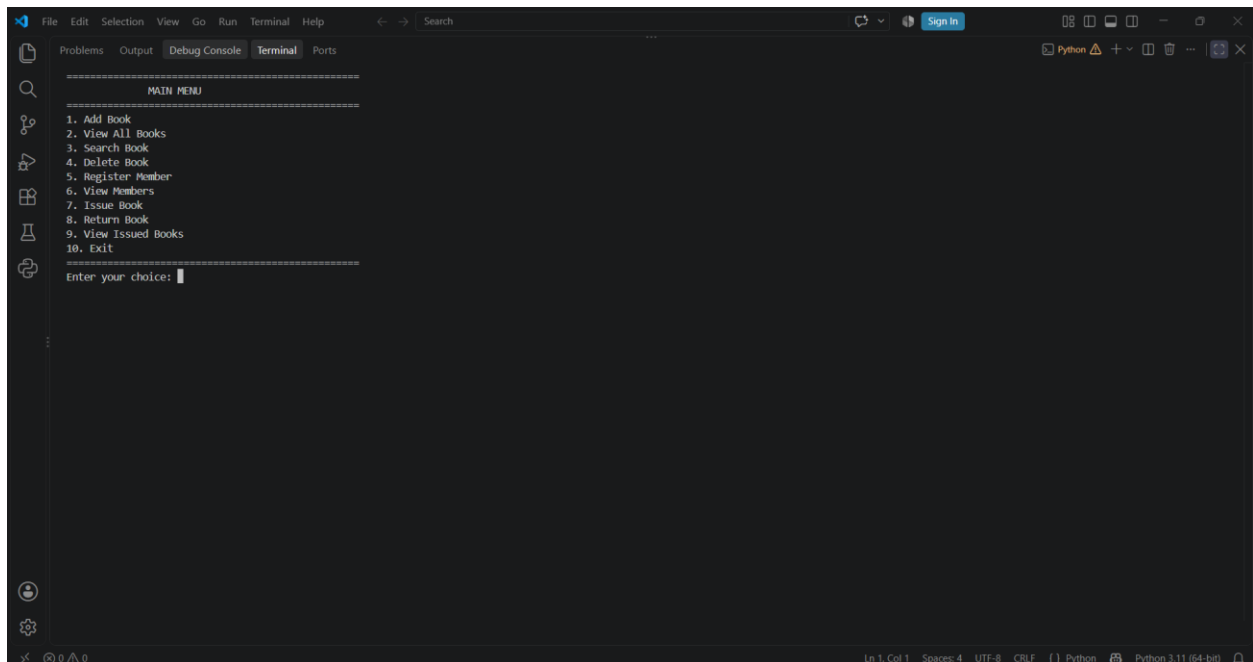
- THE RETURN DATE IS RECORDED.**
- THE BOOK QUANTITY IS INCREASED.**
- THE TRANSACTION IS COMMITTED.**

ERROR HANDLING

THE PROGRAM HANDLES INVALID USER INPUT AND MYSQL ERRORS USING EXCEPTION HANDLING.

SCREENSHOTS/RESULTS

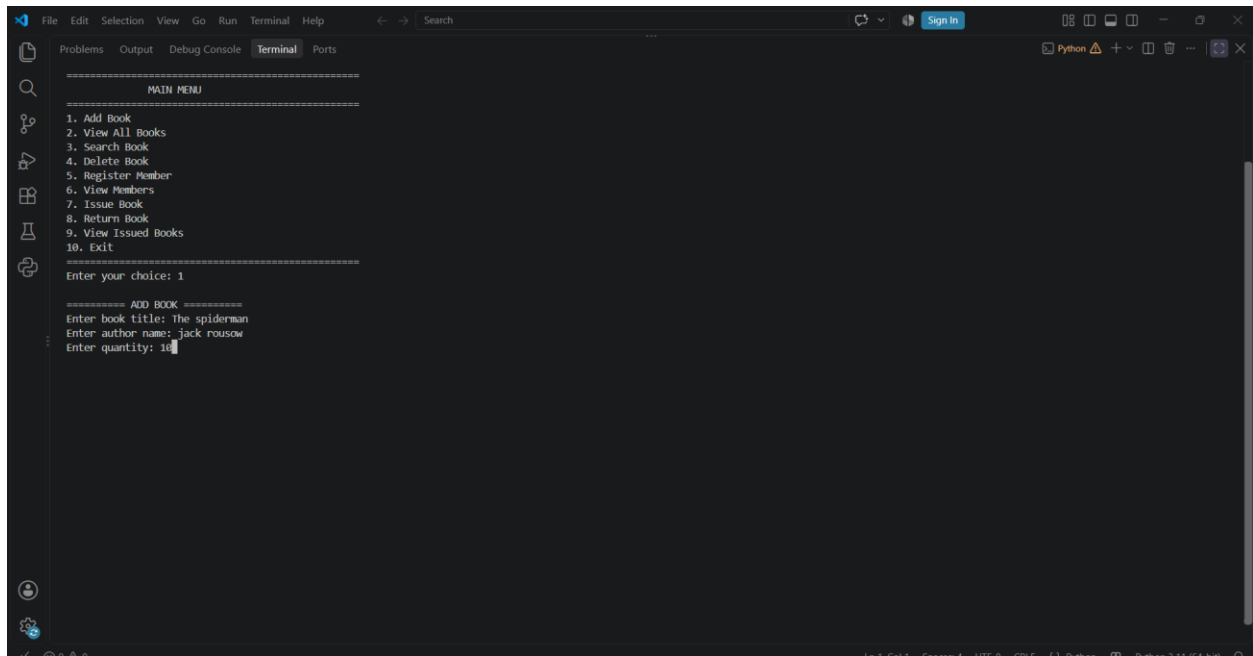
1.MAIN MENU



```
File Edit Selection View Go Run Terminal Help
Python Sign in
Problems Output Debug Console Terminal Ports
=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: |
```

The screenshot shows a Python terminal window with a dark theme. The menu is displayed with a list of 10 options. The terminal is ready for user input.

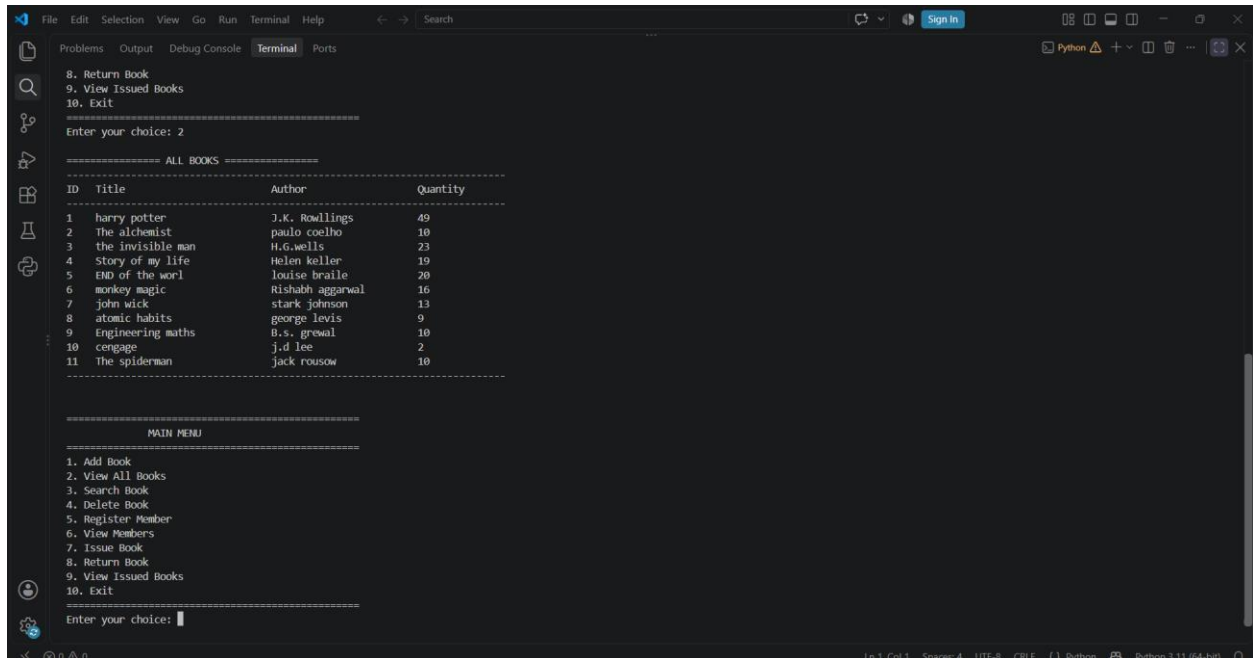
2. ADD A BOOK



```
File Edit Selection View Go Run Terminal Help
Python Sign In
Problems Output Debug Console Terminal Ports
=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: 1

===== ADD BOOK =====
Enter book title: the spiderman
Enter author name: jack rousow
Enter quantity: 10
```

3. VIEW BOOKS



```
File Edit Selection View Go Run Terminal Help
Python Sign In
Problems Output Debug Console Terminal Ports
=====
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: 2

===== ALL BOOKS =====
ID Title Author Quantity
-----
1 harry potter J.K. Rowlings 49
2 The alchemist paulo coelho 10
3 the invisible man H.G.wells 23
4 Story of my life Helen keller 19
5 END of the worl louise braile 20
6 monkey magic Rishabh aggarwal 16
7 john wick stark johnson 13
8 atomic habits george lewis 9
9 Engineering maths B.s. growal 10
10 cengage j.d lee 2
11 The spiderman jack rousow 10
=====

MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: 
```

4.REGISTER A MEMBER

```
File Edit Selection View Go Run Terminal Help
Python Sign In
Problems Output Debug Console Terminal Ports

=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: 5

===== REGISTER MEMBER =====
Enter member name: giridhar
Enter phone number: 9128453356
Member registered successfully.
Member ID: 5

=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: |
```

5.ISSUE A BOOK

```
File Edit Selection View Go Run Terminal Help
Python Sign In
Problems Output Debug Console Terminal Ports

=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: 7

===== ISSUE BOOK =====
Enter Book ID: 7
Enter Member ID: 3
Book 'john wick' issued successfully.
Issue date: 2026-09-30

=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: |
```

6.RETURN A BOOK

```
File Edit Selection View Go Run Terminal Help
Problems Output Debug Console Terminal Ports

=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: 8
===== RETURN BOOK =====
Enter Issue ID: 1
Book returned successfully.
Return date: 2026-09-30
=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: |
```

7. VIEW ISSUED BOOKS

```
File Edit Selection View Go Run Terminal Help
Problems Output Debug Console Terminal Ports

=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: 9
===== ISSUED BOOKS =====
=====
Issue ID Book Member Issue Date Return Date
=====
1 harry potter Rahul Sharma 2026-09-30 Not Returned
2 john wick ayush 2026-09-30 Not Returned
=====
MAIN MENU
=====
1. Add Book
2. View All Books
3. Search Book
4. Delete Book
5. Register Member
6. View Members
7. Issue Book
8. Return Book
9. View Issued Books
10. Exit
=====
Enter your choice: |
```

TESTING APPROACH

TESTING WAS PERFORMED TO VERIFY BOTH THE FUNCTIONALITY OF THE APPLICATION AND ITS DATABASE OPERATIONS.

FUNCTIONAL TESTING

THE FOLLOWING OPERATIONS WERE TESTED:

- ADDING BOOKS WITH VALID AND INVALID DATA.
- VIEWING AVAILABLE BOOKS.
- SEARCHING FOR EXISTING AND NON-EXISTING BOOKS.
- DELETING BOOKS.
- REGISTERING MEMBERS.
- VIEWING MEMBERS.
- ISSUING AVAILABLE BOOKS.
- ATTEMPTING TO ISSUE UNAVAILABLE BOOKS.
- RETURNING BOOKS.
- VIEWING ISSUE RECORDS.
- EXITING THE APPLICATION.

DATABASE TESTING

DATABASE OPERATIONS WERE TESTED TO VERIFY:

- CORRECT INSERTION OF RECORDS.
- CORRECT UPDATING OF QUANTITIES.
- CORRECT ISSUE AND RETURN DATES.
- CORRECT FOREIGN KEY RELATIONSHIPS.
- SUCCESSFUL TRANSACTION COMMITS.
- PROPER ROLLBACK WHEN DATABASE ERRORS OCCUR.

CHALLENGES FACED

DURING THE DEVELOPMENT OF THE PROJECT, SEVERAL CHALLENGES WERE ENCOUNTERED.

DATABASE CONNECTIVITY

CONNECTING PYTHON WITH MYSQL AND HANDLING CONNECTION ERRORS WAS AN IMPORTANT PART OF THE PROJECT.

MAINTAINING BOOK QUANTITY

THE SYSTEM NEEDED TO CORRECTLY DECREASE QUANTITY WHEN A BOOK WAS ISSUED AND INCREASE IT SOLUTION FOR MANAGING BOOKS, MEMBERS, AND BOOK TRANSACTIONS. BY COMBINING PYTHON WITH MYSQL, THE WHEN RETURNED.

PREVENTING DUPLICATE ISSUES

A VALIDATION CHECK WAS IMPLEMENTED TO PREVENT A MEMBER FROM RECEIVING THE SAME BOOK WHILE IT WAS ALREADY ISSUED TO THEM.

MAINTAINING RELATIONSHIPS

FOREIGN KEYS WERE REQUIRED TO CORRECTLY CONNECT BOOKS AND MEMBERS WITH ISSUE RECORDS.

INPUT VALIDATION

THE APPLICATION NEEDED TO HANDLE INVALID IDS, EMPTY INPUT, AND INCORRECT QUANTITIES WITHOUT CRASHING.

TRANSACTION MANAGEMENT

THE ISSUE AND RETURN OPERATIONS INVOLVE MULTIPLE DATABASE OPERATIONS, SO PROPER TRANSACTION HANDLING WAS REQUIRED TO MAINTAIN CONSISTENT DATA.

LEARNINGS & KEY TAKEAWAYS

THIS PROJECT PROVIDED PRACTICAL KNOWLEDGE ABOUT:

- **PYTHON PROGRAMMING.**
- **FUNCTIONS AND MODULAR PROGRAMMING.**
- **EXCEPTION HANDLING.**
- **MYSQL DATABASE MANAGEMENT.**
- **SQL QUERIES.**
- **CRUD OPERATIONS.**
- **PRIMARY AND FOREIGN KEYS.**
- **DATABASE RELATIONSHIPS.**
- **PYTHON-MYSQL CONNECTIVITY.**
- **TRANSACTIONS AND ROLLBACK.**
- **INPUT VALIDATION.**
- **DESIGNING A SIMPLE DATABASE-BASED APPLICATION.**

THE PROJECT ALSO IMPROVED UNDERSTANDING OF HOW SOFTWARE COMPONENTS WORK TOGETHER TO SOLVE A REAL-WORLD PROBLEM.

FUTURE ENHANCEMENTS

THE SYSTEM CAN BE FURTHER IMPROVED BY ADDING:

- **ADMIN LOGIN AND AUTHENTICATION.**
- **GRAPHICAL USER INTERFACE USING TKINTER.**
- **WEB-BASED INTERFACE USING FLASK OR DJANGO.**
- **BOOK DUE DATES.**
- **AUTOMATIC FINE CALCULATION.**
- **BOOK CATEGORIES.**
- **BOOK AVAILABILITY STATUS.**
- **MEMBER UPDATE AND DELETION FEATURES.**
- **BOOK UPDATE FUNCTIONALITY.**
- **ADVANCED SEARCH AND FILTERING.**
- **REPORTS AND STATISTICS.**
- **EXPORT RECORDS TO CSV OR PDF.**
- **EMAIL NOTIFICATIONS FOR OVERDUE BOOKS.**
- **BACKUP AND RESTORE FUNCTIONALITY.**

CONCLUSION

THE LIBRARY MANAGEMENT SYSTEM SUCCESSFULLY PROVIDES A SIMPLE COMPUTERIZED SOLUTION FOR MANAGING BOOKS, MEMBERS, AND BOOK TRANSACTIONS. BY COMBINING PYTHON WITH MYSQL, THE PROJECT DEMONSTRATES HOW A PROGRAMMING LANGUAGE CAN

INTERACT WITH A RELATIONAL DATABASE TO PERFORM REAL-WORLD MANAGEMENT TASKS.

THE SYSTEM PROVIDES A FOUNDATION THAT CAN BE EXPANDED INTO A COMPLETE DESKTOP OR WEB-BASED LIBRARY MANAGEMENT APPLICATION WITH ADDITIONAL FEATURES SUCH AS AUTHENTICATION, FINES, REPORTS, AND GRAPHICAL INTERFACES.